Phoenix Autocallable Notes on the CAC40 under a Calibrated Heston Model: Fair Coupons, Scenarios and Greeks

Elias Benguedda

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Abstract

We price single-underlying Phoenix autocallable notes, whose coupon is paid at each observation date where the index is above a coupon barrier, with or without memory of the missed coupons, on the CAC40 of 12 February 2025. The underlying follows a Heston model calibrated to the implied volatility surface (RMSE 0.42 volatility point), with a drift read from a forward curve bootstrapped from put–call parity. The note is described by a single parametrisation covering the observation frequency, coupon barrier and memory, autocall barrier and step-down, non-call period, European, American continuous or American daily capital protection, airbag, buffer, capital guarantee, range accrual coupon, duo call mechanism and look back. The notes are priced by Monte Carlo with antithetic variates and a control variate; the pricer recovers two closed-form limit cases and an exact identity with the snowball Athena. The base Phoenix (4 years, quarterly observations, memory, coupon barrier at 70%, autocall at 100%, European protection at 60%) has a fair coupon of 6.04% a year, against 10.54% for the Athena with the same dates: a unit coupon is worth 1.75 times more, since it is also paid when the index stays between 70% and 100%. We report its sensitivities to the barriers and to the market, its hedge, and the fair coupon of 20 variants. Contrary to the Athena, everything that shortens the life of the note (step-down, duo, a lower autocall barrier) raises the fair coupon, and the investor is short skew. The note is almost delta-neutral at the money, short volatility (-0.33 per vol point on a nominal of 100), short dividends (-0.32 per point of carry) and three times more exposed to the rates than the Athena.

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1 Introduction

In an AutoCallable note, the investor receives coupons well above the risk-free rate, and the note is redeemed early ("called") as soon as the index is above a reference level on an observation date. The coupon is financed by selling a down-and-in put. If the note is never called and the index ends below a protection barrier, the investor bears the loss of the index.

This report studies the **Phoenix**, whose coupon is paid at each observation date where the index is above a coupon barrier, lower than the autocall barrier, usually with a *memory* of the missed coupons. The other main family, the Athena, whose coupon is paid only when the note is called, is treated in a companion report [1]. On top of this structure, issuers combine many features: observation frequency, step-down of the autocall barrier, non-call period, European or American protection, airbag, buffer, capital guarantee, range accrual coupon, duo call mechanism, look back. We will have a look on how to price these different versions in a single framework, on real market data, and measure how each feature moves the fair coupon, the probabilities of the scenarios and the Greeks.

The model is the Heston model calibrated in [2] to the CAC40 implied volatility surface, and the Monte Carlo engine is the one used for barrier options in [3]. The scripts and notebooks are open source; Section 7 describes the code.

2 Products

2.1 Common structure

A note of nominal N on an index S has an initial fixing S_{ref} and observation dates $0 < t_1 < \cdots < t_n = T$. The performance at t_k is $P_k = S_{t_k}/S_{\text{ref}}$. All barriers are fractions of S_{ref}: autocall barriers B_k^{call}, coupon barriers B_k^{cpn} and protection barrier B^{prot}. The dates t_k with $t_k \geq t_{\text{nc}}$ are callable, t_{nc} being the end of the non-call period; the maturity is always callable. The first callable date where $P_k \geq B_k^{\text{call}}$ is the call date τ.

Redemption at maturity. If the note is never called, it is redeemed at T at

$$R_T = N\left[(1 - \mathbf{1}_{\text{KI}}) + \mathbf{1}_{\text{KI}} \max\Big(\phi,\ \min\left(1,\ P_T/K_{\text{put}} + b\right)\Big)\right], \tag{1}$$

where $\mathbf{1}_{\text{KI}}$ is the knock-in event of the protection, K_{put} the strike of the short put ϕ the capital floor and b the buffer ($b = 0$ by default). The knock-in event is

- *European protection*: $P_T < B^{\text{prot}}$;
- *American protection*: the index touches $B^{\text{prot}} S_{\text{ref}}$ at any time before maturity, continuously or on daily closes.

The standard note has $K_{\text{put}} = 1$ and $\phi = 0$: below the barrier, the investor receives NP_T and the redemption jumps from N to NB^{prot} at the barrier. The *airbag* note has $K_{\text{put}} = B^{\text{prot}}$: the redemption is continuous at the barrier and equal to NP_T/B^{prot} below it. A *capital guarantee* sets $\phi > 0$. A *buffer* sets $b > 0$ with $B^{\text{prot}} = 1 - b$: the investor only loses below $1 - b$, one for one, without loss jump. Figure 1 draws the three profiles.

2.2 Phoenix

The Phoenix pays at each observation date $t_k \leq \tau$ (and $t_k \leq T$) the coupon

$$N\, c\, (t_k - t_{k-1})\, \mathbf{1}_{\{P_k \geq B_k^{\text{cpn}}\}} \tag{2}$$

and, with memory, the coupons of all the previous dates since the last payment. When the note is called, it is redeemed at N. With $B_k^{\text{cpn}} = 0$, the coupon is unconditional; with $B_k^{\text{cpn}} = B_k^{\text{call}}$

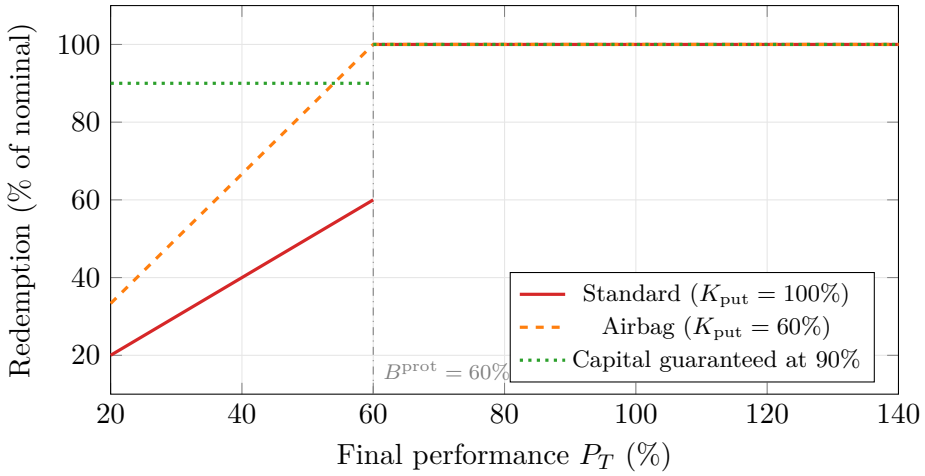


Figure 1: Redemption at maturity of a note that was never called, European protection at 60% (the last coupon is paid on top of it if $P_T \geq B_n^{\text{cpn}}$).

and memory, a coupon is paid only at the call date and its amount is $c\,t_k$: the Phoenix is then exactly a snowball Athena (Table 3 checks it numerically).

Optional features. Three features change the call or the coupon:

- *range accrual*: the coupon of each period only accrues on the dates of the period where $L \leq S_t/S_{\text{ref}} \leq U$; it is still paid only if $P_k \geq B_k^{\text{cpn}}$;
- *duo*: a second call level $B_k^{\text{duo}} < B_k^{\text{call}}$; if $B_k^{\text{duo}} \leq P_k < B_k^{\text{call}}$, the note is called and the coupon of the date is multiplied by α;
- *look back*: S_{ref} is the minimum (or the average) of the index over $[0, t_{\text{lb}}]$, path by path, instead of the initial fixing, all the barriers being fractions of this reference.

2.3 Parametrisation

Table 1 lists the parameters of the class `Phoenix`; every variant of Section 5 is one of these parameters changed.

Feature	Parameter	Values
Maturity, initial fixing, nominal	`maturity`, `initial_fixing`, `nominal`	years, index level, 100
Observation dates	`frequency` or `observation_dates`	monthly to annual, or any list ending at T
Coupon rate	`coupon_rate`	per annum, fraction of the nominal
Coupon condition	`coupon_barrier`, `memory`	fraction of S_{ref} (scalar or one per date), True / False
Autocall barrier	`autocall_barrier`, `step_down`	fraction of S_{ref}, decrease per observation
Non-call period	`first_call_date`	years
Capital protection	`protection_barrier`, `protection_type`	fraction of S_{ref}; `european`, `american`, `american_discrete`
Airbag, guarantee	`put_strike`, `floor`	fraction of S_{ref}, fraction of the nominal
Buffer	`buffer`	fraction of S_{ref}, with `protection_barrier` $= 1 - b$
Range accrual	`accrual_range`	(L, U), fractions of S_{ref}
Duo	`duo_barrier`, `duo_coupon_ratio`	fraction of S_{ref}, fraction α of the coupon
Look back	`lookback`, `lookback_type`	years, `min` or `average`

Table 1: Parameters of the Phoenix (`Products/Phoenix/phoenix.py`).

3 Market Data and Model

The data are the CAC40 option chain of 12 February 2025 (spot $S_0 = 8042.19$, 142 quotes on 13 expiries from 9 days to 4.86 years) and the EURIBOR 6M zero-coupon curve. The forward curve $F(0,t)$ is bootstrapped from put–call parity with a piecewise constant borrow-plus-dividend yield $q(t)$ [2]; its integral reaches 0.136 at 4.86 years, so dividends lower the 4-year forward by about 11% with respect to a zero carry. Under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = \partial_t \ln F(0,t)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{3}$$

with the parameters of Table 2. The notes are struck at the money: $S_{\mathrm{ref}} = S_0 = 8042.19$, and S_{ref} stays fixed when the Greeks bump the market. All the dates of the 4-year notes lie before the last listed expiry (4.86 years).

v_0	θ	κ	ξ	ρ	IV RMSE
0.0197	0.0438	1.308	0.482	-0.605	0.421 vol pts

Table 2: Heston parameters calibrated on the CAC40 surface of 12 February 2025 [2].

4 Pricing Method

4.1 Simulation

The paths of (S, v) are simulated with an Euler full truncation scheme on a weekly grid (52 steps a year, 208 steps for 4 years), the drift over each step being $\ln(F(0,t_{i+1})/F(0,t_i))$, so that $\mathbb{E}[S_t] = F(0,t)$ at every date. $N = 50{,}000$ paths with antithetic variates are used (seed 42); prices and standard errors are computed on the 25,000 independent pair averages. The observation dates are rounded to the grid; with weekly steps, every monthly, quarterly, semi-annual and annual date falls within half a week of its contractual date.

4.2 Cash flows and protection

For each path, the note is followed through its observation dates: at each date, the coupon is paid if the performance is above the coupon barrier (with the unpaid coupons if the note has a memory), then the note is called if the performance is above the autocall barrier. Each cash flow is discounted with $D(0,t_k) = e^{-r(t_k)t_k}$. The knock-in indicator of (1) is

- *European*: $\mathbf{1}_{\{P_T < B^{\text{prot}}\}}$;
- *American continuous*: replaced by its conditional probability given the simulated path, $1 - \prod_i (1 - p_i)$, where $p_i = \exp\big(-2\ln(S_{t_i}/H)\ln(S_{t_{i+1}}/H)/(v_{t_i}\Delta t)\big)$ is the probability that the Brownian bridge of $\ln S$ touches $H = B^{\text{prot}} S_{\text{ref}}$ between two grid dates [8];
- *American daily*: the barrier is checked on a daily grid (252 steps a year).

4.3 Variance reduction

Besides antithetic variates, the discounted European put struck at $B^{\text{prot}} S_{\text{ref}}$, whose Heston price is known semi-analytically (Lewis formula), is used as a control variate with an optimal coefficient estimated on the paths.

4.4 Fair coupon and scenario statistics

On a fixed set of paths, the call dates do not depend on the coupon, and every cash flow is affine in the coupon rate c: the price is $V(c) = V(0) + c\,(V(1) - V(0))$, and the coupon that prices the note at par is

$$c^\star = \frac{N - V(0)}{V(1) - V(0)}, \tag{4}$$

obtained with two valuations. It excludes the issuer margin and funding spread. The same paths give the probability of a call at each date, the probability to reach maturity, the probability of a capital loss ($\mathbf{1}_{\text{KI}} = 1$ and $P_T < K_{\text{put}}$), the expected loss at maturity, the expected life $\mathbb{E}[\min(\tau, T)]$ and the expected number of coupons.

4.5 Greeks

The Greeks are central finite differences with common random numbers [7], on a nominal of 100: Delta and Gamma (spot $\pm 1\%$, carry kept), Vega and Volga ($\sqrt{v_0}$ and $\sqrt{\theta}$ shifted by ± 1 vol point), Vanna, Rho (± 10bp on the zero curve, per $+1\%$), Dividend (± 10bp on $q(t)$, per $+1\%$) and Theta (dates moved by one day). Delta is given per index point: multiply by $S_0/100 = 80.4$ for the value change of a 1% move. Each Greek is reported with its Monte Carlo standard error.

4.6 Validation

Two limit cases have a closed form and one an exact counterpart (Table 3). In a flat Black–Scholes market ($S_0 = S_{\text{ref}} = 100$, $r = 3\%$, $q = 1\%$, $\sigma = 20\%$, $T = 2$, $N = 100$, Heston with $v_0 = \theta = \sigma^2$ and $\xi = 10^{-4}$, 10^5 paths):

- a Phoenix never called (autocall barrier out of reach) with unconditional quarterly coupons and a European protection at $B = 60$ is worth the sum of its discounted coupons plus $ND(T) - \frac{N}{S_{\text{ref}}}\left[\text{Put}(B) + (S_{\text{ref}} - B)\,\text{DigitalPut}(B)\right]$;
- a Phoenix called for sure at t_1 is worth $N(1 + c\,t_1)D(t_1)$.

On the CAC40, the Phoenix with memory and $B^{\text{cpn}} = B^{\text{call}}$ must have exactly the price of the snowball Athena with the same dates.

Case	**Reference**	**Monte Carlo**	**Std error**	**Error / std error**
Phoenix never called, unconditional 6% coupon (flat)	104.2404	104.2515	0.0141	0.79
Phoenix called for sure at $t_1 = 0.25$, $c = 6\%$ (flat)	100.742	100.742	0	exact
Phoenix with memory, $B^{\text{cpn}} = B^{\text{call}}$ vs snowball Athena (CAC40)	98.7669	98.7669	0.0585	identical paths

Table 3: Validation of the pricer.

5 Phoenix : Base case

Let us consider, under the risk-neutral measure $\mathbb{Q}$, the general case of a Phoenix in the base case, on a nominal of 1, with N observation dates, a coupon c per period, a coupon barrier B_c, an autocall barrier B_{ac} and a protection barrier B_p, with $B_p \leq B_c \leq B_{ac}$. As for the Athena, we denote by

$$E_i = \bigcap_{j=1}^{i-1} \{X_j < B_{ac}\}, \qquad i = 1, \dots, N, \qquad E_1 = \Omega, \tag{5}$$

the event that the product has not been called before date t_i. The event $\{\tau = t_i\}$ of an early redemption at date $t_i < T$ is then $E_i \cap \{X_i \geq B_{ac}\}$, and $\{\tau = T\} = E_N$. With memory, the coupon paid at t_i covers all the periods since the last payment: it is $c\, m_i$, with

$$m_i = i - \max(\{0\} \cup \{j < i : X_j \geq B_c\}), \tag{6}$$

and $m_i = 1$ without memory.

5.1 Price of the Phoenix

For a coupon c, the price of the Phoenix at inception is the sum of its discounted coupons and of its discounted redemption,

$$P_0 = \mathbb{E}^{\mathbb{Q}}\Big[\sum_{t_i \leq \tau} D(t_i)\, c\, m_i\, \mathbf{1}_{\{X_i \geq B_c\}} + D(\tau)\, \Pi_\tau\Big], \tag{7}$$

which, writing the autocall events explicitly, expands as

$$\begin{aligned} P_0 = &\sum_{i=1}^{N} D(t_i)\, c\, \mathbb{E}^{\mathbb{Q}}\big[m_i\, \mathbf{1}_{E_i \cap \{X_i \geq B_c\}}\big] \\ &+ \sum_{i=1}^{N-1} D(t_i)\, \mathbb{Q}(E_i \cap \{X_i \geq B_{ac}\}) \\ &+ D(T)\, \mathbb{Q}(E_N \cap \{X_N \geq B_p\}) \\ &+ D(T)\, \mathbb{E}^{\mathbb{Q}}\big[X_N\, \mathbf{1}_{E_N \cap \{X_N < B_p\}}\big]. \end{aligned} \tag{8}$$

The first term is the strip of conditional coupons, the second corresponds to an early redemption at date t_i, the third to the repayment of the notional at maturity, and the last one to the capital loss scenario. Contrary to the Athena, the coupon is not attached to the redemption: it is paid as long as the note is alive and the index above B_c, including in scenarios that end with a capital loss.

5.2 Linearity in the coupon and embedded put

The price of the Phoenix is also affine in the coupon: the coupon affects the amounts paid, never the redemption time τ nor the dates where a coupon is paid, which depend on the path of the underlying alone. On the event E_N, the terminal payoff can be rewritten, since $B_p < 1$, as

$$\Pi_T = 1 + c\, m_N\, \mathbf{1}_{\{X_N \geq B_c\}} - (1 - X_N)^+\, \mathbf{1}_{\{X_N < B_p\}}, \tag{9}$$

where the last term is again the down-and-in put with strike 100% and barrier B_p observed at maturity, sold by the investor. Substituting (9) into (8) yields the decomposition

$$P_0 = \underbrace{\mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big]}_{\text{notional}} + \underbrace{c \sum_{i=1}^{N} D(t_i)\, \mathbb{E}^{\mathbb{Q}}\big[m_i\, \mathbf{1}_{E_i \cap \{X_i \geq B_c\}}\big]}_{B\ \text{(coupons)}} - \underbrace{D(T)\, \mathbb{E}^{\mathbb{Q}}\Big[(1 - X_N)^+\, \mathbf{1}_{\{X_N < B_p\}}\, \mathbf{1}_{E_N}\Big]}_{\text{PDI}^{ac}}, \tag{10}$$

that is

$$P_0 = A + c\,B, \qquad A = \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] - \mathrm{PDI}^{ac}. \tag{11}$$

The zero-coupon part A does not depend on the coupon condition: for the same dates and the same autocall and protection barriers, *the Phoenix and the Athena have the same A*. They only differ through the value B of a unit coupon. Two limit cases bound it:

- with $B_c = 0$ (unconditional coupon), $m_i = 1$ and $B = \sum_i D(t_i)\,\mathbb{Q}(E_i)$, a strip of coupons knocked out at the call;
- with $B_c = B_{ac}$ and memory, $X_j < B_c$ for every $j < i$ on E_i, so that $m_i = i$ and B is exactly the B of the snowball Athena.

5.3 Fair coupon

As for the Athena, the fair coupon for a target price $P^\star$ (1, or $1 - m$ with an issuer margin m) is

$$\boxed{c^\star = \frac{P^\star - A}{B} = \frac{P^\star - \mathbb{E}^{\mathbb{Q}}\big[D(\tau)\big] + \mathrm{PDI}^{ac}}{B}} \tag{12}$$

and a single Monte Carlo simulation provides A and B. Since A is shared with the Athena of the same dates, the ratio of the two fair coupons is the inverse ratio of the coupon values,

$$\frac{c^\star_{\text{Phoenix}}}{c^\star_{\text{Athena}}} = \frac{B_{\text{Athena}}}{B_{\text{Phoenix}}}. \tag{13}$$

On the base case, the quarterly snowball Athena pays 10.54% and the Phoenix 6.04%: a unit coupon is worth 1.75 times more in the Phoenix, because it is also paid in the scenarios where the index stays between B_c and B_{ac}.

5.4 Remarks on the embedded put and on the coupons

The put PDI^{ac} is the same as in the Athena: knocked out by the autocall, bounded by the European down-and-in put, which decomposes into a vanilla put struck at the barrier and a digital put,

$$(1 - X_N)^+\,\mathbf{1}_{\{X_N < B_p\}} = (B_p - X_N)^+ + (1 - B_p)\,\mathbf{1}_{\{X_N < B_p\}}. \tag{14}$$

The coupons add a second exposure to the shape of the distribution: each coupon is a digital call struck at B_c, knocked out by an earlier call. Without memory, $B = \sum_i D(t_i)\,\mathbb{Q}(E_i \cap \{X_i \geq B_c\})$; the memory can only increase it, up to $\sum_i i\,D(t_i)\,\mathbb{Q}(E_i \cap \{X_i \geq B_c\})$. Both strips are sensitive to the skew around B_c.

5.5 Structuration :

From the investor's perspective, the basic Phoenix note is composed of:

- a long position in an *autocallable zero-coupon note* with maximal maturity T, which repays the notional at the first observation date t_i such that $X_i \geq B_{ac}$, and at maturity otherwise;
- a long position in a *strip of digital coupons* paying c (or $c\,m_i$ with memory) at each date t_i where $X_i \geq B_c$, knocked out when the product is autocalled;
- a short position in a *down-and-in put* (PDI) with strike 100% and barrier B_p observed at maturity, which is knocked out whenever the product is autocalled.

5.6 Sensitivity analysis :

We will now look at the different sensitivities of the product and what are the impacts of a move on certain parameters on c. We are now thinking from the investor point of view and with an already contracted note.

Volatility

The investor is short the PDI, whose price increases with volatility: he is short vega. The digital coupons, struck at $B_c = 70\%$ below the spot, also lose value when the volatility rises, since the index is less likely to stay above the coupon barrier. The base note loses 0.33 per vol point.

If the product is not contracted yet, a higher volatility makes the put more expensive, and the fair coupon c increases.

Protection barrier

If the protection barrier is higher, then it is more probable to cross it and it reduces the price of the note. The effect on the coupon c is that it is increased: 5.16% at 50%, 6.04% at 60%, 6.85% at 70%.

Recall Barrier

If the recall barrier is increased, it is less probable to be recalled before maturity: the investor keeps the put longer, which should raise the fair coupon c. Contrary to the Athena, a longer life also means more coupons, since the Phoenix pays its coupon at every date where the index is above B_c. Each point of coupon is then worth more to the investor, and this effect dominates above 95%: the fair coupon is 5.46% with an autocall barrier at 90%, 6.10% at 95%, 6.04% at 100% and 5.31% at 110%. Conversely, a step-down of the autocall barrier, which shortens the coupon stream, raises the fair coupon.

Coupon barrier

If the coupon barrier is increased, fewer coupons are paid and the fair coupon c increases: 5.43% at 50%, 6.04% at 70%, 6.67% at 80%, and 5.13% for an unconditional coupon.

Memory

The memory pays back the missed coupons as soon as the index comes back above B_c: it is worth 0.37 point of coupon at a coupon barrier of 70% (6.04% instead of 6.41% without memory), and 2.41 points at 90%.

Dividend

If the underlying asset increases its dividend, the forward decreases: the put is more valuable, the calls less likely, and the fair coupon c increases. The investor does not receive the dividends. But a lower forward also delays the call, so that more coupons are paid: the exposure is about half of the Athena one (-0.32 per point of carry against -0.69 for the quarterly Athena).

Spread

The issuer funds itself at $r + s$, s being its credit spread, and discounts the cash flows of the note at this rate. A higher spread lowers $\mathbb{E}[D(\tau)]$ and increases the funding benefit $1 - \mathbb{E}[D(\tau)]$, which is given back to the investor through a higher coupon. Since a unit coupon is worth more in a Phoenix than in an Athena (B is larger), the same funding benefit buys fewer points of coupon: the coupon of the Phoenix is about half as sensitive to the spread.

Risk free rate

As for the Athena, a higher rate raises the forwards, which calls the note earlier and lowers the put, but it also lowers the discount factors and increases the funding benefit of the issuer. For the Phoenix, a third effect adds to the second one: an earlier call removes coupons. The coupon c increases with the rates, more than for the Athena.

Maturity

A longer maximal maturity leaves more time for the knock-in of the paths that are not called, and the put is more valuable; it also adds coupon dates, which makes each point of coupon worth more. The first effect dominates: the fair coupon increases with the maturity, at a decreasing rate since most notes are called during the first year.

Skew

The skew acts on the Phoenix in two ways. The put is a vanilla put struck at the barrier plus a digital put, priced by the skew: a steeper skew makes it more expensive. And a negative skew makes the distribution of the index more left-skewed with a higher median: the note is called earlier, so fewer coupons are paid. For the Athena, the earlier call offsets the more expensive put; for the Phoenix, both effects raise the coupon. The investor of a Phoenix is short skew.

Correlation

On a single underlying, the only correlation is the one between the index and its volatility, which drives the skew. For a worst-of Phoenix on several underlyings, the investor is long correlation: a lower correlation makes it more likely that one of the underlyings ends below the protection barrier or below the coupon barrier, and less likely that all of them are above the autocall barrier. The put is more valuable and the coupons less likely, so the coupon c increases. Worst-of notes are outside the scope of this report (Section 8).

5.7 Pricing : A numerical example

Let us now consider a synthetic Phoenix written on the CAC 40 index (dividends not reinvested). Its initial level is fixed on 12 February 2025, the date of the market data, and its maximal maturity is 4 years. The underlying is observed quarterly: at each observation date, a coupon of 1.5% (6% per annum) is paid if the index closes at or above the coupon barrier $B_c = 70\%$ of its initial level, together with the coupons missed since the last payment (memory), and the product is redeemed early if the index closes at or above 100% of its initial level. If the product has not been called, the investor receives at maturity the last coupon if the index is at or above the coupon barrier, the notional if it is at or above the protection barrier $B_p = 60\%$, and the notional reduced by the full decline of the index otherwise. The base case is priced without the credit spread of the issuer, whose impact is studied with the other market parameters below.

5.7.1 Model

The underlying follows the Heston model of Section 3, calibrated on the CAC 40 implied volatility surface of 12 February 2025, with the forward curve bootstrapped from put–call parity. The paths are simulated on a weekly grid with 50,000 paths (Section 4); a daily grid is used for the daily observations and the daily monitored protection. The base case has the same dates and barriers as the base Athena of the companion report [1], and it is priced on the same paths. All the figures and tables of this section come from the notebook `Products/Phoenix/phoenix_synthetic_pricing.ipynb`.

5.7.2 Pricing

For the purpose of the pricing of this product, we use the script `phoenix.py`[1] to build the product and `monte_carlo.py`[2] to simulate the different paths. In this first basic case, we have the following result:

		Base case
Contract	Maturity (years)	4
	Number of observations	16
	Observation frequency (years)	0.25
	Coupon (% p.a.)	6
	First coupon barrier (%)	70
	Last coupon barrier (%)	70
	Memory	Yes
	First autocall barrier (%)	100
	Last autocall barrier (%)	100
	Step-down per observation (%)	0
	First call date (years)	0.25
	Protection barrier (%)	60
	Protection type	European
	Redemption if knocked-in	Standard
	Capital guaranteed (%)	0
	Buffer (%)	0
	Range accrual (%)	–
	Duo barrier (%)	–
	Duo coupon ratio	–
	Look-back	–
	Nominal	100
Pricing	Price (% nominal)	99.96
	95% confidence interval (±)	0.10
	Fair coupon (% p.a.)	6.04
Scenarios	Probability to reach maturity (%)	13.29
	Probability of a capital loss (%)	5.22
	Expected loss at maturity (% nominal)	3.05
	Expected life (years)	1.03
	Expected number of coupons	3.28
	Expected coupon paid (% nominal)	5.23

Table 4: Contract features, pricing results and scenario statistics of the base-case Phoenix.

We also plotted the probability of a call at each observation date.

The call probabilities are exactly those of the Athena with the same autocall barrier: the coupon does not change the call rule. 55.4% of the notes are called after one quarter, and 86.7% before maturity. What changes is the way the coupon is paid. The Phoenix pays 3.28 coupons on average, for a total of 5.23% of the nominal, against 3.01% for the Athena: it also pays in the scenarios where the index stays between 70% and 100%, where the Athena pays nothing. This is why the coupon needed to price the note at par, 6.04%, is much lower than the Athena's 10.54%. The investor is still short the same down-and-in put, with a loss jump from 100 to 60 at the protection barrier.

[1] https://github.com/benguedalauzonelias/portfolio/tree/main/Products/Phoenix
[2] https://github.com/benguedalauzonelias/portfolio/tree/main/Equity_Models

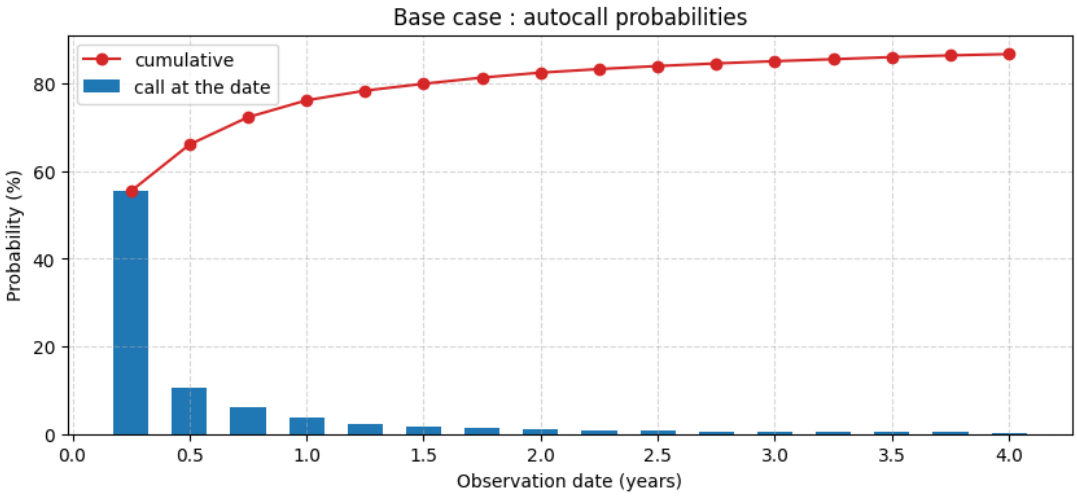


Figure 2: Base Phoenix: probability of a call at each observation date, and cumulated.

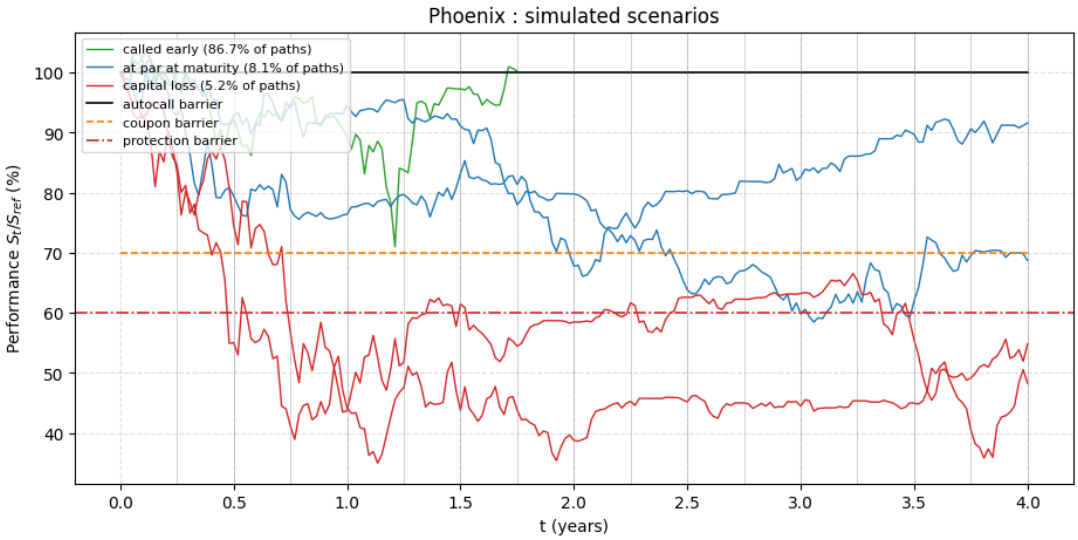


Figure 3: Base Phoenix: simulated paths of the performance, grouped by outcome, with the coupon, autocall and protection barriers.

5.7.3 Sensitivities

We are now interested in how the different parameters impact the price of a basic Phoenix product. As for the Athena, the price after the emission and the fair coupon $c^\star$ move in opposite directions. Each parameter is moved alone, all the others being those of the base case, on the same paths (common random numbers).

Protection barrier

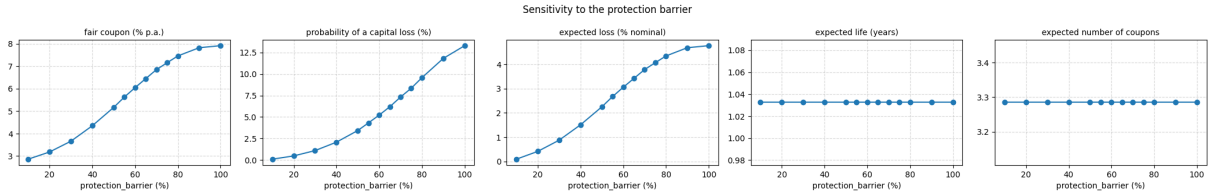


Figure 4: Fair coupon, probability of a capital loss, expected loss, expected life and expected number of coupons as functions of the protection barrier.

Protection barrier (%)	Fair coupon (% p.a.)	Probability of a capital loss (%)	Expected loss (% nominal)
10	2.86	0.10	0.09
20	3.18	0.48	0.41
30	3.66	1.09	0.87
40	4.35	2.07	1.50
50	5.16	3.43	2.25
55	5.63	4.32	2.67
60	**6.04**	**5.22**	**3.05**
65	6.45	6.19	3.42
70	6.85	7.32	3.78
75	7.15	8.34	4.07
80	7.45	9.57	4.34
90	7.82	11.81	4.68
100	7.91	13.29	4.77

Table 5: Fair coupon and loss statistics as a function of the protection barrier (base case in bold). The expected life (1.03 years) and the expected number of coupons (3.28) do not depend on the protection barrier.

A higher protection barrier exposes the investor more often to a capital loss: the put he sells is more valuable, and the fair coupon increases, from 5.16% at 50% to 7.45% at 80%. The loss statistics are exactly those of the Athena, since the two notes share the same put; the fair coupon moves less, because each point of coupon is worth 1.75 times more in the Phoenix. The calls and the coupons do not move: the protection only matters at maturity.

Autocall barrier

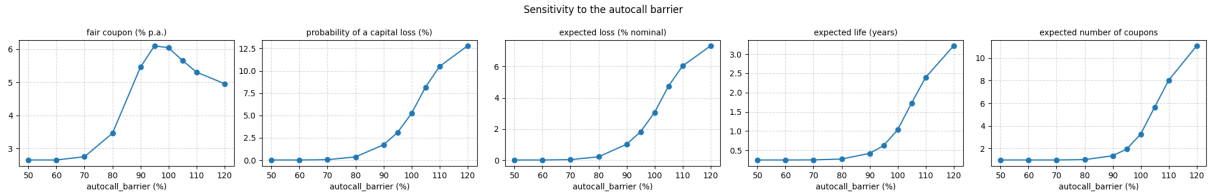


Figure 5: Fair coupon, loss statistics, expected life and expected number of coupons as functions of the autocall barrier.

Autocall barrier (%)	Fair coupon (% p.a.)	Probability of a capital loss (%)	Expected loss (% nominal)	Expected life (years)	Expected number of coupons
50	2.65	0.00	0.00	0.25	1.00
70	2.75	0.04	0.02	0.25	1.00
80	3.46	0.35	0.21	0.28	1.03
90	5.46	1.70	1.01	0.42	1.37
95	6.10	3.08	1.82	0.62	1.96
100	**6.04**	**5.22**	**3.05**	**1.03**	**3.28**
105	5.66	8.16	4.75	1.73	5.67
110	5.31	10.48	6.04	2.39	8.04
120	4.96	12.79	7.33	3.21	11.07

Table 6: Fair coupon, loss statistics, expected life and coupons as a function of the autocall barrier (base case in bold).

Here the Phoenix and the Athena part ways. A higher autocall barrier makes the call less likely: the note lives longer and more paths are exposed to the put (probability of a loss from 1.70% at 90% to 10.48% at 110%). For the Athena, which pays a single coupon at the call, the fair coupon rises from 6.66% to 12.69%. For the Phoenix, a longer life also means more coupon dates: 8.04 coupons are paid on average at 110% against 3.28 at 100%, so each point of coupon is worth much more. The fair coupon first increases, up to 6.10% at 95%, then decreases (5.31% at 110%, 4.96% at 120%): above 95%, the additional coupons outweigh the additional put. Below 70%, the note is called at the first date on every path, and the fair coupon is the three-month rate, 2.65%, as for the Athena.

Coupon barrier

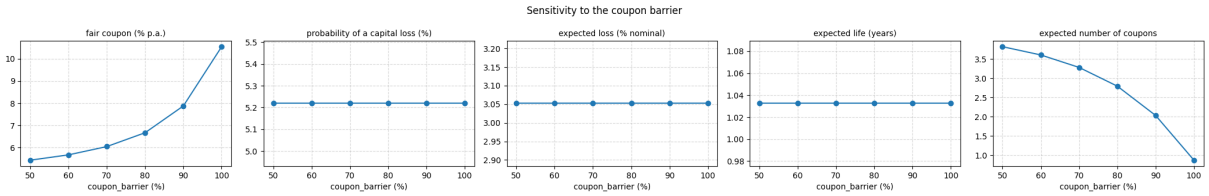


Figure 6: Fair coupon and scenario statistics as functions of the coupon barrier.

Coupon barrier (%)	Fair coupon (% p.a.)	Expected number of coupons
50	5.43	3.83
60	5.67	3.61
70	**6.04**	**3.28**
80	6.67	2.80
90	7.87	2.03
100	10.54	0.87

Table 7: Fair coupon and expected number of coupons as a function of the coupon barrier (base case in bold). The calls and the losses do not depend on the coupon barrier.

A higher coupon barrier pays fewer coupons, so the fair coupon increases, slowly up to 70% and steeply above: from 80% to 100%, the expected number of coupons falls from 2.80 to 0.87. With memory, a coupon barrier equal to the autocall barrier gives back exactly the snowball Athena: the coupons are only paid at the call, all together, and the fair coupon is the Athena's, 10.54%. Figure 7 compares the coupon barrier curve with and without memory, and the protection barrier curve with a European and an American protection.

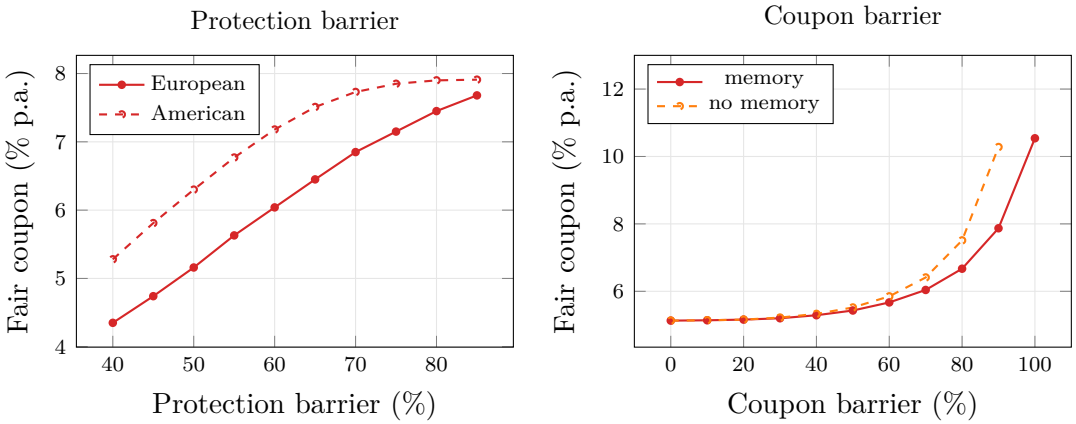


Figure 7: Phoenix: fair coupon as a function of the barriers (other features of the base case unchanged; American: continuous monitoring). Without memory, the fair coupon reaches 23.94% at a coupon barrier of 100% (off scale).

Market parameters

The note is now kept unchanged (coupon 6%, barriers of the base case) and the market moves (Table 8).

Volatility. One volatility point costs about 0.33 on a nominal of 100, as for the Athena, and adds about 0.40 point of fair coupon (from 4.94% at −3 points to 7.24% at +3 points). The investor is short the put, whose value increases with the volatility, and the coupons, digital calls struck at 70%, are slightly less likely (3.37 to 3.19 coupons).

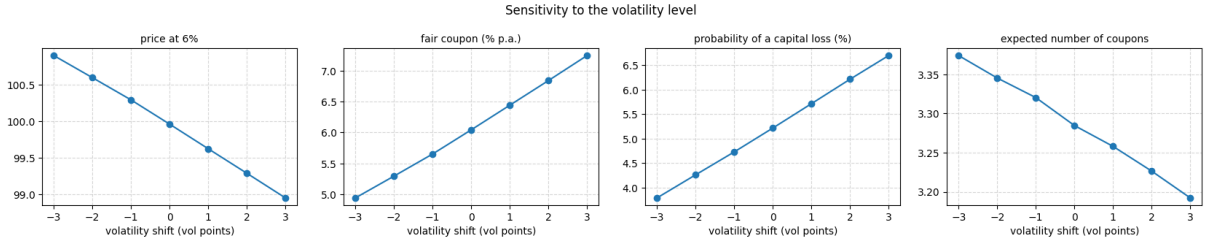


Figure 8: Base Phoenix: price at 6%, fair coupon, probability of a capital loss and expected number of coupons as functions of a parallel shift of the volatility level.

Dividends. A higher dividend yield lowers the forwards: the put is more valuable and the note is called later. For the Athena, both effects raise the coupon. For the Phoenix, the later call also adds coupons (3.10 coupons at −1%, 3.48 at +1%), which partly offsets the put: the fair coupon only moves from 5.73% to 6.40%, against 9.38% to 12.02% for the Athena.

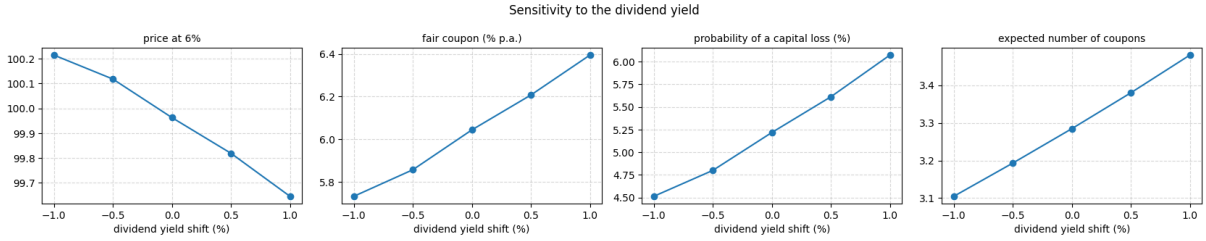


Figure 9: Base Phoenix as a function of a parallel shift of the borrow + dividend yield.

Risk-free rate. Higher rates raise the forwards and the discounting. The put is cheaper, but the funding benefit of the issuer is larger and the note is called earlier, with fewer coupons: the price at 6% falls from 100.64 at −1% to 99.35 at +1% and the fair coupon rises from 5.30% to 6.82%, 0.76 point per point of rate (0.61 for the Athena). The Phoenix is a more rate-sensitive product than the Athena.

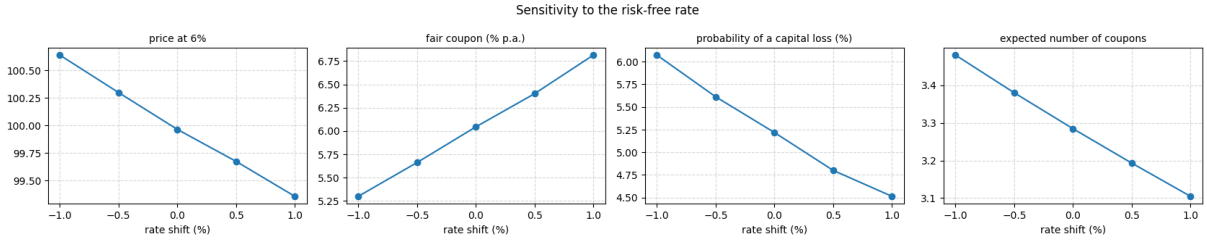


Figure 10: Base Phoenix as a function of a parallel shift of the zero-coupon curve.

Issuer spread. Discounting the cash flows at $r + s$ costs about 0.46 per 50bp and adds 0.55 point of fair coupon per 50bp (6.06% at $s = 0$, 8.24% at $s = 200$bp, control variate switched off), about half of the Athena sensitivity: the funding benefit is the same, but it is spread over a larger value of the unit coupon B.

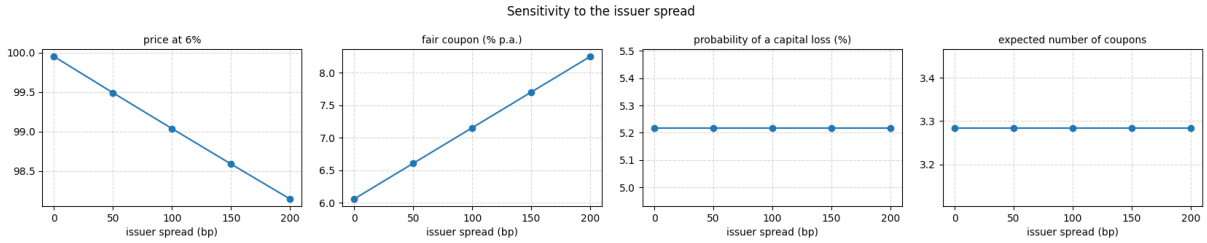


Figure 11: Base Phoenix as a function of the credit spread of the issuer.

Maturity. The fair coupon increases with the maturity at a decreasing rate: 4.21%, 5.33%, 5.77%, 6.04% and 6.15% from 1 to 5 years. A longer maturity makes the put more valuable (probability of a loss from 1.67% to 5.62%) and adds coupon dates (1.98 to 3.50 coupons), the first effect dominating.

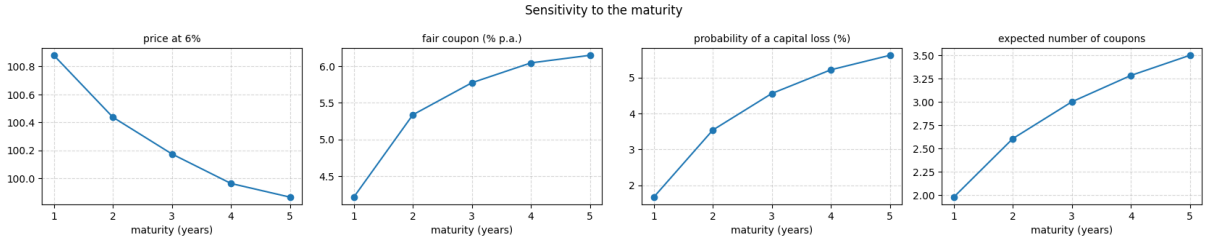


Figure 12: Same Phoenix with a maximal maturity from 1 to 5 years.

Skew. The spot-variance correlation ρ is moved alone, from -0.9 to 0. A more negative ρ makes the low strikes more expensive (4-year put at 60%: 2.39% of S_0 at $\rho = -0.9$ against 1.54% at $\rho = 0$) and the call earlier (60.0% at the first date against 46.9%). For the Athena, these two effects offset each other. For the Phoenix, the earlier call removes coupons (2.82 coupons at $\rho = -0.9$ against 4.24 at $\rho = 0$), so that both effects raise the coupon: the fair coupon goes from 5.18% at $\rho = 0$ to 6.40% at $\rho = -0.9$. The Phoenix investor is short skew, the Athena investor almost neutral.

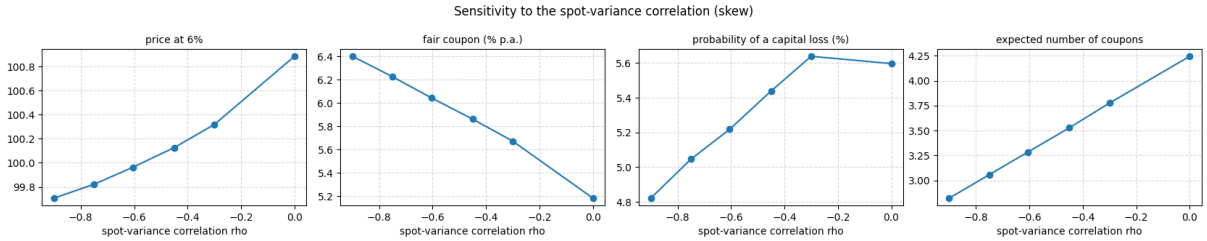


Figure 13: Base Phoenix as a function of the spot-variance correlation ρ of the Heston model.

	Scenario	Price at 6%	Fair coupon (% p.a.)	P(call t_1) (%)	P(loss) (%)	E[coupons]
Volatility	−3 vol pts	100.90	4.94	56.82	3.79	3.37
	base	99.96	6.04	55.43	5.22	3.28
	+3 vol pts	98.95	7.24	54.17	6.70	3.19
Dividend yield	−1%	100.22	5.73	56.95	4.51	3.10
	+1%	99.65	6.40	53.88	6.07	3.48
Rates	−1%	100.64	5.30	53.88	6.07	3.48
	+1%	99.35	6.82	56.95	4.51	3.10
Issuer spread	0bp	99.95	6.06	55.43	5.22	3.28
	100bp	99.04	7.15	55.43	5.22	3.28
	200bp	98.15	8.24	55.43	5.22	3.28
Maturity	1 year	100.88	4.21	55.43	1.67	1.98
	2 years	100.44	5.33	55.43	3.54	2.61
	5 years	99.87	6.15	55.43	5.62	3.50
Correlation ρ	−0.9	99.71	6.40	59.97	4.82	2.82
	0	100.89	5.18	46.89	5.60	4.24

Table 8: Base Phoenix under moved market parameters (coupon 6%, nominal 100). Base case: price 99.96, fair coupon 6.04%.

5.7.4 Hedging

From the desk's point of view, the issuer is short the note: it is long the down-and-in put sold by the investor and short the strip of conditional coupons and the autocall feature. Table 9 gives its hedge at inception for a nominal of 1 million euros (10,000 notes of 100).

Risk of the note (nominal 100)	**Value**	**Hedge of the issuer** (nominal 1M)
Delta	0.00059 per index point	buy 5.9 index units ($\approx$ 47,000 EUR of futures)
Vega	-0.335 per vol point	long 3,350 EUR per vol point: sell 173 puts 4 years, strike 60%
Dividend	-0.325 per $+1\%$	long 3,250 EUR per point of yield: sell dividend futures
Rho	-0.599 per $+1\%$	long 5,990 EUR per point of rate: receive fixed on a swap

Table 9: Base Phoenix: Greeks of the note and hedge of the issuer at inception (vega of the 4-year put struck at 60%: 19.3 per vol point and per unit of index).

At inception, the Phoenix is almost delta-neutral: it is equivalent to 4.7% of its nominal invested in the index, against 37.8% for the Athena. A rise of the index reduces the short put, but it also brings the call forward and stops a coupon stream that is worth more than its cost. The hedge is much less stable than this number suggests (Figure 14): when the index falls, the call becomes unlikely and the note behaves more and more like a short put, so its delta rises to 0.0044 at 85% of the initial level, 0.0097 at 75% and 0.0213 at the protection barrier. The issuer must buy index in a falling market, from almost nothing. The vega of the note is most negative, about -1.13 per vol point, between 75% and 80% of the initial level, just above the coupon barrier, where both the coupons and the knock-in are most uncertain. The rate risk is three times that of the Athena: the issuer pays a regular coupon stream, which it hedges with swaps. Close to maturity, the jumps of the payoff at the coupon barrier and at the protection barrier create large delta and gamma around these two levels (pin risk).

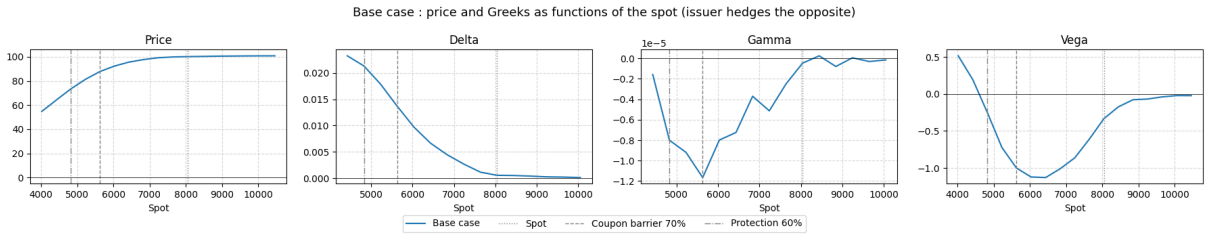


Figure 14: Base Phoenix: price, delta, gamma and vega of the note as functions of the spot (30,000 paths). The issuer holds the opposite of these Greeks.

5.8 Features of Phoenix products

We have seen the basic Phoenix product, its construction, its sensitivities and its hedging. We now add the features that issuers combine with it, one at a time, all the other terms being those of the base case. Each feature is one parameter of the class `Phoenix` (Table 1).

5.8.1 Exercise frequency

The coupon and autocall barriers can be observed every year, every semester, every month or every day instead of every quarter, the coupon of a period being c times its length.

		Base case	**Annual**	**Semi-annual**	**Monthly**	**Daily**
Contract	Number of observations	16	4	8	48	1008
	Observation frequency (years)	0.25	1	0.5	1/12	1/252
Pricing	Price (% nominal)	99.96	100.62	100.22	99.91	99.94
	95% confidence interval (±)	0.10	0.11	0.11	0.08	0.04
	Fair coupon (% p.a.)	6.04	5.63	5.82	6.19	6.52
Scenarios	Probability to reach maturity (%)	13.29	25.70	18.61	7.77	1.80
	Probability of a capital loss (%)	5.22	9.06	6.97	3.13	0.78
	Expected loss at maturity (% nominal)	3.05	5.27	4.08	1.83	0.46
	Expected life (years)	1.03	2.06	1.48	0.59	0.13
	Expected number of coupons	3.28	1.67	2.38	5.53	25.97
	Expected coupon paid (% nominal)	5.23	10.39	7.49	2.98	0.67

Table 10: Effect of the observation frequency of the coupon and autocall barriers, all other parameters being those of the base case.

More frequent observations call the note earlier (expected life from 2.06 years to 0.13 year) and reduce the probability of a loss (from 9.06% to 0.78%), but they also stop the coupon stream earlier: the coupon paid falls from 10.39% to 0.67% of the nominal. The fair coupon increases with the frequency, from 5.63% (annual) to 6.52% (daily), in the same direction as for the Athena but over a much narrower range (9.41% to 11.74% for the Athena): since the Phoenix also pays coupons when the note is not called, its coupon depends less on the timing of the call.

5.8.2 Memory

With memory, the coupons missed because the index was below the coupon barrier are paid at the next date where it is back above. Without memory, they are lost for good.

		Base case	**No memory**
Contract	Memory	Yes	No
Pricing	Price (% nominal)	99.96	99.67
	95% confidence interval (±)	0.10	0.10
	Fair coupon (% p.a.)	6.04	6.41
Scenarios	Probability to reach maturity (%)	13.29	13.29
	Probability of a capital loss (%)	5.22	5.22
	Expected loss at maturity (% nominal)	3.05	3.05
	Expected life (years)	1.03	1.03
	Expected number of coupons	3.28	3.28
	Expected coupon paid (% nominal)	5.23	4.93

Table 11: Phoenix without memory: a missed coupon is lost for good.

The coupon dates are the same, but the amounts paid are smaller without memory (4.93% of the nominal instead of 5.23%): the fair coupon rises from 6.04% to 6.41%. The memory is worth 0.37 point of coupon here, and much more with a higher coupon barrier (2.41 points at 90%, Figure 7), where coupons are missed more often.

5.8.3 Step-down barrier

The autocall barrier decreases by 1% at each observation date, from 100% to 85%.

		Base case	**Step-down 1%**
Contract	Last autocall barrier (%)	100	85
	Step-down per observation (%)	0	1
Pricing	Price (% nominal)	99.96	99.89
	95% confidence interval (±)	0.10	0.09
	Fair coupon (% p.a.)	6.04	6.16
Scenarios	Probability to reach maturity (%)	13.29	8.12
	Probability of a capital loss (%)	5.22	4.44
	Expected loss at maturity (% nominal)	3.05	2.62
	Expected life (years)	1.03	0.87
	Expected number of coupons	3.28	2.73
	Expected coupon paid (% nominal)	5.23	4.38

Table 12: Step-down autocall barrier, decreasing by 1% per quarterly observation from 100% to 85%.

The step-down calls the notes that drift slightly below their initial level: the probability to reach maturity falls from 13.29% to 8.12% and the probability of a loss from 5.22% to 4.44%. For the Athena, this lowers the fair coupon from 10.54% to 8.24%. For the Phoenix, it slightly raises it, to 6.16%: the earlier calls stop the coupon stream (2.73 coupons instead of 3.28), and this loss of coupons outweighs the smaller put. A step-down makes a Phoenix safer, but not cheaper.

5.8.4 Airbag

The put sold by the investor is struck at the protection barrier: below 60%, the investor receives NP_T/B^{prot} instead of NP_T, and the redemption is continuous at the barrier.

		Base case	**Airbag**
Contract	Put strike (%)	100	60
Pricing	Price (% nominal)	99.96	101.28
	95% confidence interval (±)	0.10	0.07
	Fair coupon (% p.a.)	6.04	4.49
Scenarios	Probability to reach maturity (%)	13.29	13.29
	Probability of a capital loss (%)	5.22	5.22
	Expected loss at maturity (% nominal)	3.05	1.61
	Expected life (years)	1.03	1.03
	Expected number of coupons	3.28	3.28
	Expected coupon paid (% nominal)	5.23	5.23

Table 13: Airbag: the put is struck at the protection barrier (60%), all other parameters being those of the base case.

The scenarios and the coupons are those of the base case; only the amount lost changes. The expected loss is halved (1.61% instead of 3.05%) and the fair coupon falls to 4.49%. As for the Athena, the airbag removes the digital part of the put, hence the loss jump and most of the pin risk at the protection barrier.

5.8.5 Buffer

With a buffer of 10%, the investor only loses below 90%, one for one: the redemption is $N \min(1, P_T + 10\%)$ below the protection barrier $B^{\text{prot}} = 90\%$.

		Base case	**Buffer 10%**
Contract	Protection barrier (%)	60	90
	Buffer (%)	0	10
Pricing	Price (% nominal)	99.96	99.54
	95% confidence interval (±)	0.10	0.08
	Fair coupon (% p.a.)	6.04	6.55
Scenarios	Probability to reach maturity (%)	13.29	13.29
	Probability of a capital loss (%)	5.22	11.81
	Expected loss at maturity (% nominal)	3.05	3.50
	Expected life (years)	1.03	1.03
	Expected number of coupons	3.28	3.28
	Expected coupon paid (% nominal)	5.23	5.23

Table 14: Buffer of 10%: the investor only loses below 90%, one for one, all other parameters being those of the base case.

The losses start as soon as the index ends below 90%: the probability of a loss rises from 5.22% to 11.81% and the fair coupon from 6.04% to 6.55%. But there is no more loss jump, and the largest losses are reduced by 10 points.

5.8.6 Guaranteed capital

The redemption at maturity is floored at 90% or 100% of the nominal.

		Base case	**Guaranteed 90%**	**Guaranteed 100%**
Contract	Capital guaranteed (%)	0	90	100
Pricing	Price (% nominal)	99.96	102.26	102.74
	95% confidence interval (±)	0.10	0.04	0.03
	Fair coupon (% p.a.)	6.04	3.33	2.77
Scenarios	Probability to reach maturity (%)	13.29	13.29	13.29
	Probability of a capital loss (%)	5.22	5.22	0.00
	Expected loss at maturity (% nominal)	3.05	0.52	0.00
	Expected life (years)	1.03	1.03	1.03
	Expected number of coupons	3.28	3.28	3.28
	Expected coupon paid (% nominal)	5.23	5.23	5.23

Table 15: Capital guarantee: the redemption at maturity is floored at 90% or 100% of the nominal.

With a full guarantee, the put disappears and the coupons are only financed by the funding benefit of the issuer: the fair coupon falls to 2.77% (4.83% for the Athena, whose single coupon is paid less often). With 90% guaranteed, the expected loss is divided by six and the fair coupon is 3.33%. A capital-guaranteed Phoenix is close to a callable bond paying a conditional coupon, financed by the rates.

5.8.7 Coupon range accrual

The coupon of each quarter only accrues on the weekly dates of the quarter where the index is between 80% and 120%; it is still paid only if the index is above the coupon barrier at the observation date.

		Base case	**Range accrual 80-120%**
Contract	Range accrual (%)	–	[80, 120]
Pricing	Price (% nominal)	99.96	99.02
	95% confidence interval (±)	0.10	0.10
	Fair coupon (% p.a.)	6.04	7.41
Scenarios	Probability to reach maturity (%)	13.29	13.29
	Probability of a capital loss (%)	5.22	5.22
	Expected loss at maturity (% nominal)	3.05	3.05
	Expected life (years)	1.03	1.03
	Expected number of coupons	3.28	3.28
	Expected coupon paid (% nominal)	5.23	4.25

Table 16: Range accrual: the coupon of each quarter only accrues on the weekly dates where the index is between 80% and 120%.

The range accrual has a much larger effect than on the Athena (10.99% instead of 10.54%): the coupon paid falls from 5.23% to 4.25% of the nominal and the fair coupon rises from 6.04% to 7.41%. The Phoenix pays a coupon for every quarter of its life, so every week spent outside the range costs a coupon, whereas the Athena only pays at the call, usually after one quarter.

5.8.8 Duo mechanism

Two call levels: above 100%, the note is called with the coupon of the date; between 90% and 100%, it is also called, but the coupon of the date (with the memory) is halved.

		Base case	**Duo 90% / 50%**
Contract	Duo barrier (%)	–	90
	Duo coupon ratio	–	0.5
Pricing	Price (% nominal)	99.96	99.89
	95% confidence interval (±)	0.10	0.06
	Fair coupon (% p.a.)	6.04	6.35
Scenarios	Probability to reach maturity (%)	13.29	3.13
	Probability of a capital loss (%)	5.22	1.70
	Expected loss at maturity (% nominal)	3.05	1.01
	Expected life (years)	1.03	0.42
	Expected number of coupons	3.28	1.37
	Expected coupon paid (% nominal)	5.23	1.87

Table 17: Duo mechanism: call above 100%, and also between 90% and 100% with half of the coupon of the date.

The note is called on almost every path and very early (probability to reach maturity 3.13%, expected life 0.42 year), so the probability of a loss falls to 1.70%. For the Athena, this lowers the fair coupon (8.86%). For the Phoenix, the coupon stream is cut short (1.37 coupons instead of 3.28) and the fair coupon rises to 6.35%: as for the step-down, the duo makes the Phoenix safer but not cheaper.

5.8.9 Look back

The reference level S_{ref} is the minimum of the index over the first month: every barrier, including the coupon barrier, is a fraction of a lower level on the paths that fell during the first month.

		Base case	**Look back min 1m**
Contract	Look back	–	min over 1 month
Pricing	Price (% nominal)	99.96	100.20
	95% confidence interval (±)	0.10	0.09
	Fair coupon (% p.a.)	6.04	5.69
Scenarios	Probability to reach maturity (%)	13.29	9.03
	Probability of a capital loss (%)	5.22	3.57
	Expected loss at maturity (% nominal)	3.05	2.09
	Expected life (years)	1.03	0.78
	Expected number of coupons	3.28	2.53
	Expected coupon paid (% nominal)	5.23	4.01

Table 18: Look back: the reference level is the minimum of the index over the first month.

The look back calls the note earlier (expected life 0.78 year) and lowers the probability of a loss (3.57% instead of 5.22%). It removes coupons through the earlier call (2.53 coupons instead of 3.28) but also makes each coupon more likely, since the coupon barrier is lower on the paths that fell. Here the reduction of the loss dominates: the fair coupon falls to 5.69%.

5.8.10 Protection type

The protection can be American: the capital is at risk as soon as the index touches 60% at any time, continuously or on daily closes.

		Base case	**American**	**American (daily)**
Contract	Protection type	European	Continuous	Daily closes
Pricing	Price (% nominal)	99.96	99.00	99.07
	95% confidence interval (±)	0.10	0.10	0.10
	Fair coupon (% p.a.)	6.04	7.18	7.11
Scenarios	Probability to reach maturity (%)	13.29	13.29	13.23
	Probability of a capital loss (%)	5.22	9.22	8.98
	Expected loss at maturity (% nominal)	3.05	4.11	4.06
	Expected life (years)	1.03	1.03	1.03
	Expected number of coupons	3.28	3.28	3.28
	Expected coupon paid (% nominal)	5.23	5.23	5.22

Table 19: American protection at 60%, observed continuously (Brownian bridge) or on daily closes.

The probability of a loss rises from 5.22% to 9.22% and the fair coupon from 6.04% to 7.18% (7.11% with a daily monitoring). The American protection costs 1.14 points of coupon, against 1.99 for the Athena: the same put premium is spread over a larger unit coupon value.

5.8.11 Non-call period

The note cannot be called during the first year; the coupons are still paid during this period.

		Base case	**Non-call 1 year**
Contract	First call date (years)	0.25	1
Pricing	Price (% nominal)	99.96	100.86
	95% confidence interval (±)	0.10	0.10
	Fair coupon (% p.a.)	6.04	5.46
Scenarios	Probability to reach maturity (%)	13.29	22.06
	Probability of a capital loss (%)	5.22	8.42
	Expected loss at maturity (% nominal)	3.05	4.92
	Expected life (years)	1.03	1.89
	Expected number of coupons	3.28	6.21
	Expected coupon paid (% nominal)	5.23	9.81

Table 20: Non-call period of one year (the coupons are still paid during this period).

The note lives longer (1.89 years) and is more often exposed to the put (probability of a loss 8.42%), but it also pays many more coupons (6.21 instead of 3.28): the fair coupon falls to 5.46%. A non-call period is the way to guarantee the investor a minimum number of coupons.

5.8.12 Unconditional coupon

With a coupon barrier at 0, the coupon is paid at every date until the call or the maturity, whatever the level of the index.

		Base case	**Unconditional coupon**
Contract	Coupon barrier (%)	70	0
Pricing	Price (% nominal)	99.96	100.87
	95% confidence interval (±)	0.10	0.08
	Fair coupon (% p.a.)	6.04	5.13
Scenarios	Probability to reach maturity (%)	13.29	13.29
	Probability of a capital loss (%)	5.22	5.22
	Expected loss at maturity (% nominal)	3.05	3.05
	Expected life (years)	1.03	1.03
	Expected number of coupons	3.28	4.13
	Expected coupon paid (% nominal)	5.23	6.19

Table 21: Unconditional coupon: the coupon is paid at every date until the call, whatever the level of the index.

The coupon is paid more often (4.13 coupons instead of 3.28), and the fair coupon falls to 5.13%. The note is then a callable fixed-rate bond combined with the short down-and-in put, the simplest structure of the family.

5.9 All the forms of the Phoenix

Table 22 gathers all the variants of this section, sorted by fair coupon, and Figure 15 draws their fair coupons.

Variant	Price at 6%	Fair coupon (% p.a.)	P(maturity) (%)	P(loss) (%)	E[loss] (% nom.)	E[life] (y)	E[coupons] (number)
Guaranteed 100%	102.74	2.77	13.29	0.00	0.00	1.03	3.28
Guaranteed 90%	102.26	3.33	13.29	5.22	0.52	1.03	3.28
Airbag	101.28	4.49	13.29	5.22	1.61	1.03	3.28
Unconditional coupon	100.87	5.13	13.29	5.22	3.05	1.03	4.13
Autocall 110%	101.40	5.31	37.56	10.48	6.04	2.39	8.04
Non-call 1 year	100.86	5.46	22.06	8.42	4.92	1.89	6.21
Annual observation	100.62	5.63	25.70	9.06	5.27	2.06	1.67
Look back min 1m	100.20	5.69	9.03	3.57	2.09	0.78	2.53
Semi-annual observation	100.22	5.82	18.61	6.97	4.08	1.48	2.38
Base case	99.96	6.04	13.29	5.22	3.05	1.03	3.28
Step-down 1%	99.89	6.16	8.12	4.44	2.62	0.87	2.73
Monthly observation	99.91	6.19	7.77	3.13	1.83	0.59	5.53
Duo 90% / 50%	99.89	6.35	3.13	1.70	1.01	0.42	1.37
No memory	99.67	6.41	13.29	5.22	3.05	1.03	3.28
Daily observation	99.94	6.52	1.80	0.78	0.46	0.13	25.97
Buffer 10%	99.54	6.55	13.29	11.81	3.50	1.03	3.28
American protection (daily)	99.07	7.11	13.23	8.98	4.06	1.03	3.28
American protection	99.00	7.18	13.29	9.22	4.11	1.03	3.28
Range accrual 80-120%	99.02	7.41	13.29	5.22	3.05	1.03	3.28
Protection 80%	98.77	7.45	13.29	9.57	4.34	1.03	3.28
Coupon barrier 90%	98.78	7.87	13.29	5.22	3.05	1.03	2.03

Table 22: All the Phoenix variants of this section, sorted by fair coupon (Heston, 50,000 paths, coupon 6% p.a.). Each variant changes one feature of the base case.



Figure 15: Fair coupon of each Phoenix variant (the base case in red).

Reading the table.

- *Capital protection.* As for the Athena, the guarantee (2.77% at 100%, 3.33% at 90%) and the airbag (4.49%) reduce the loss, while the American protection (7.18%), a higher barrier (7.45% at 80%) and the buffer (6.55%) make it more likely.
- *Coupon condition.* The coupon barrier, the memory and the range accrual act directly on the coupons: unconditional coupon 5.13%, no memory 6.41%, range accrual 7.41%, coupon barrier at 90% 7.87%.

- *Life of the note.* Contrary to the Athena, everything that shortens the life of the Phoenix stops its coupon stream: the step-down (6.16%), the duo (6.35%) and more frequent observations (6.52% daily) raise the fair coupon, while a higher autocall barrier (5.31% at 110%) and a non-call period (5.46%) lower it.

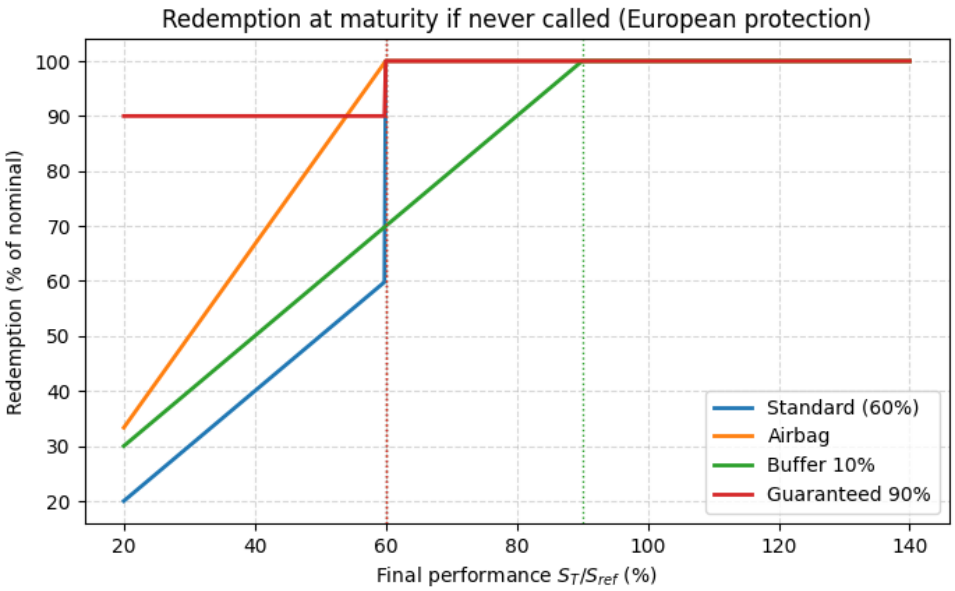


Figure 16: Redemption at maturity of a Phoenix never called: standard, airbag, buffer and capital guaranteed at 90%.

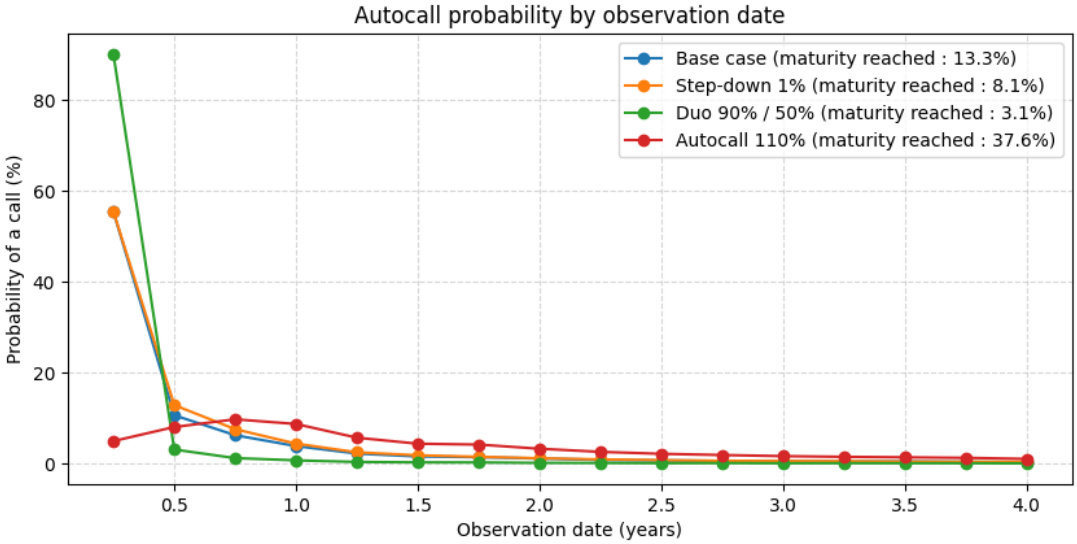


Figure 17: Probability of a call at each observation date: base case, step-down, duo and autocall barrier at 110%.

5.10 Greeks

Table 23 gives the Greeks of six Phoenix variants (nominal 100, coupon 6%). Gamma and Volga are not significant at this number of paths (base case: Gamma $-1.2 \times 10^{-6} \pm 4.3 \times 10^{-6}$, Volga -0.005 ± 0.024); the base case standard errors are 0.00017 on the Delta, 0.012 on the Vega, 0.044 on the Rho and on the Dividend, and 0.0014 on the Theta.

Phoenix	**Price**	**Delta** (per point)	**Delta 1%** (per 1% move)	**Vega** (per vol pt)	**Vanna** ($\times 10^4$)	**Rho** (per 1%)	**Dividend** (per 1%)	**Theta** (per day)
Base case	99.96	0.00059	0.047	−0.335	6.13	−0.599	−0.325	0.0050
American protection	99.00	0.00229	0.184	−0.394	5.72	−0.448	−0.437	0.0039
Step-down 1%	99.89	0.00046	0.037	−0.270	3.37	−0.579	−0.198	0.0032
Airbag	101.28	−0.00151	−0.121	−0.224	5.38	−0.875	−0.101	0.0026
Guaranteed 90%	102.26	−0.00297	−0.239	−0.086	4.37	−1.073	0.057	0.0021
No memory	99.67	0.00106	0.085	−0.365	6.27	−0.560	−0.357	0.0053

Table 23: Greeks of Phoenix variants (nominal 100, coupon 6%).

The Phoenix is almost delta-neutral at the money (0.047 for a 1% move against 0.378 for the Athena): a rise of the index reduces the short put, but it also calls the note earlier, which stops a coupon stream worth more than its cost. When the put is reduced (airbag) or mostly removed (guaranteed 90%), the second effect wins and the delta becomes negative. For the same reason, the guaranteed note has a positive dividend exposure: a higher dividend yield delays the call and adds coupons, and this outweighs the small put that is left. The Phoenix is short volatility (−0.33 per vol point, close to the Athena), short dividends (−0.32 per point, half of the Athena) and much more exposed to the rates (−0.60 per point against −0.21), since it pays a regular coupon stream. Without memory, the note is more sensitive to the spot and to the volatility, since its coupons depend more on the level of each date.

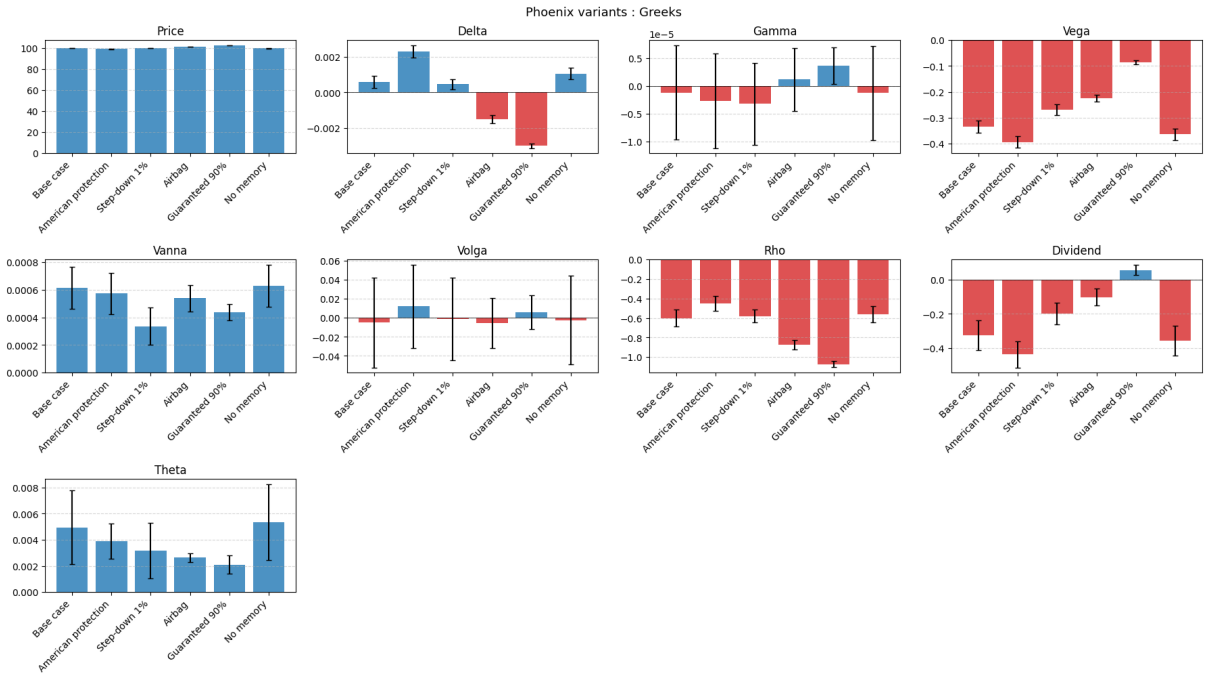


Figure 18: Greeks of the Phoenix variants with 95% confidence intervals.

Sensitivities to the Heston parameters. Per unit of parameter, the base Phoenix has $\partial V/\partial v_0 = -40.4 \pm 12.2$ and $\partial V/\partial \theta = -44.9 \pm 7.4$, while $\partial V/\partial \kappa = -0.19 \pm 0.21$, $\partial V/\partial \xi = 1.4 \pm 1.0$

and $\partial V/\partial\rho = 0.9 \pm 0.7$ are not significant at this number of paths. Contrary to the Athena, which depends mostly on the initial variance v_0, the Phoenix is as sensitive to the long-term variance θ: its coupon stream lasts longer than the single coupon of the Athena. The sensitivity to ρ, not significant for a small bump, is clear over the wider range of the skew scenarios above.

6 Phoenix versus Athena

The two notes of this report and of its companion [1] share the same dates, the same autocall barrier and the same protection: they have the same call dates, the same put and the same zero-coupon part A, and only differ by the way the coupon is paid. Table 24 compares them.

	Athena (snowball)	**Phoenix** (memory, CB 70%)
Fair coupon (% p.a.)	10.54	6.04
Probability of a call at the first date (%)	55.43	55.43
Expected life (years)	1.03	1.03
Probability of a capital loss (%)	5.22	5.22
Expected loss (% of nominal)	3.05	3.05
Expected coupon paid at 6% (% of nominal)	3.01	5.23
Delta (per 1% move)	0.378	0.047
Vega (per vol point)	−0.315	−0.335
Rho (per 1%)	−0.210	−0.599
Dividend (per 1% of carry)	−0.689	−0.325
Fair coupon (% p.a.) of some variants		
Autocall barrier 110%	12.69	5.31
Step-down 1%	8.24	6.16
Duo 90% / 50%	8.86	6.35
Non-call 1 year	8.94	5.46
Daily observation	11.74	6.52
Range accrual 80–120%	10.99	7.41
American protection	12.53	7.18
Rates −1% / +1%	9.95 / 11.16	5.30 / 6.82
Correlation $\rho = -0.9$ / 0	10.51 / 10.33	6.40 / 5.18

Table 24: Base cases compared (4 years, quarterly observations, autocall at 100%, European protection at 60%, same paths); Athena figures from the companion report [1].

The Athena concentrates its coupon in the scenarios where the index is up at an observation date: its fair coupon is higher, its delta eight times larger, and its value depends on the timing of the call, a later call paying a larger snowball coupon. The Phoenix pays regularly as long as the index stays above 70%: it is a stream of conditional coupons, more stable for the investor and worth more per point of coupon. Everything that lengthens the life of the note works in its favour: a higher autocall barrier or a non-call period lower its fair coupon, a step-down, a duo or more frequent observations raise it, whereas they lower the Athena coupon. The Phoenix is also short skew and much more exposed to the rates, while the Athena is more exposed to the spot and to the dividends. With a coupon barrier equal to the autocall barrier and memory, the two products coincide (10.54%), which is the identity of Section 2.

7 Implementation

7.1 Code

- `Products/Phoenix/phoenix.py`: the self-contained class `Phoenix` (contract, `path_payoff`, `fair_coupon`, `statistics`, and `features` / `report` for one table with the terms, the pricing and the scenarios);
- `Equity_Models/monte_carlo.py`: `MonteCarloPricer`, which simulates the paths and calls `path_payoff` for products with several cash flows;
- `Equity_Models/Heston.py`, `Equity_Models/forward_curve.py`: calibrated model and forward curve;
- `Equity_Models/greeks.py`: Greeks with common random numbers, including the Dividend Greek;
- `Products/market.py`: CAC40 market data and calibrated Heston model (`cac40_heston`);
- `Products/notes_analysis.py`: comparison of variants and plots;
- notebooks `Products/Phoenix/phoenix_pricer.ipynb` and `Products/Phoenix/phoenix_synthetic_pricing.ipynb` (every number of Section 5);
- `Reports/Phoenix/make_figures.py`: extracts the figures of this report from the executed notebook.

```
from market import cac40_heston
from phoenix import Phoenix
from Equity_Models.monte_carlo import MonteCarloPricer
from Equity_Models.greeks import Greeks

mkt = cac40_heston()
pricer = MonteCarloPricer(n_paths=50_000, steps_per_year=52, seed=42)
note = Phoenix(4.0, mkt.S0, 0.06, coupon_barrier=0.7, memory=True, frequency
    =0.25,
               autocall_barrier=1.0, protection_barrier=0.6, protection_type="
                   european")
print(pricer.price(mkt.heston, note))              # price and standard error
print(note.fair_coupon(pricer, mkt.heston))        # coupon p.a. at par
print(note.report(pricer, mkt.heston))             # terms, pricing and scenario
    statistics
print(Greeks(pricer).compute(mkt.heston, note, Greeks.ORDER))
```

7.2 Cost

With N paths and M steps, the simulation costs $\mathcal{O}(NM)$ in time and stores S and v, i.e. $16(M+1)N$ bytes: 167 MB for $N = 50{,}000$ and the weekly grid of 4 years, 807 MB on a daily grid. Following the note through its n observation dates costs $\mathcal{O}(nN)$, the American continuous protection $\mathcal{O}(NM)$. Measured times (one CPU core): about 0.3 s for a valuation including the simulation, and a few hundredths of a second for the price, the fair coupon and the statistics of a variant on cached paths. The whole notebook `phoenix_synthetic_pricing.ipynb` (calibration, 20 variants, 3 barrier sweeps, 6 market sweeps, the Greeks of six notes and a spot profile) runs in about 1.5 minutes.

8 Limitations and Next Steps

- *Single underlying.* Most autocalls sold today are worst-of on two to four indices or stocks; they need a multi-asset model and a correlation calibration.
- *Model.* The value of an autocall depends on the forward skew and on the joint dynamics of spot and volatility between observation dates; a pure Heston model with constant parameters is not the market standard, which is a local stochastic volatility model calibrated to the whole surface [6].
- *Fair coupon.* It prices the note at par without issuer margin; the credit spread of the issuer is only studied as a parallel discounting spread, without default modelling. A real issue price is below par by the margin.
- *Discretisation.* Euler full truncation with the Feller condition violated ($2\kappa\theta/\xi^2 = 0.49$), weekly grid with the variance frozen over each week for the Brownian bridge.
- *Greeks.* Gamma and Volga are not significant with 50,000 paths: the payoff jumps at the coupon, autocall and protection barriers.
- *Dividends.* Continuous dividend yield bootstrapped from the forwards; real notes are exposed to discrete dividends and to dividend risk, which the Dividend Greek only measures for a parallel shift.

9 Conclusion

A single parametrisation covers the Phoenix with all its usual features, and a Monte Carlo pricer under a calibrated Heston model recovers two exact limit cases within one standard error and the exact identity with the snowball Athena. On the CAC40 of 12 February 2025, the base 4-year quarterly Phoenix has a fair coupon of 6.04% a year; its zero-coupon part is the same as the Athena one, and its lower coupon only reflects that a unit coupon is worth 1.75 times more. The protection and the coupon condition (barrier, memory, range accrual) are the main drivers of the fair coupon, as for the Athena. But the Phoenix investor is long a stream of conditional coupons whose value grows with the life of the note: everything that shortens it (step-down, duo, more frequent observations) raises the fair coupon, everything that lengthens it (higher autocall barrier, non-call period) lowers it, the opposite of the Athena. The note is almost delta-neutral at the money, short volatility, short skew, short dividends (half of the Athena) and three times more exposed to the rates; for the issuer, the delta to buy grows from almost nothing to 0.02 per point when the index falls to the protection barrier.

A Notation

Symbol	Meaning
N, S_{ref}, T	Nominal, initial fixing, maturity
t_k, $\delta_k = t_k - t_{k-1}$	Observation dates and accrual periods
$P_k = S_{t_k}/S_{\text{ref}}$	Performance at t_k
B_k^{call}, B_k^{cpn}, B^{prot}	Autocall, coupon and protection barriers (fractions of S_{ref})
m_i	Number of coupon periods paid at t_i (memory)
c, $c^\star$	Coupon rate per annum, fair coupon
K_{put}, ϕ, b	Strike of the short put, capital floor, buffer
τ	Call date
$\mathbf{1}_{\text{KI}}$	Knock-in event of the protection

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